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CMAQ Model Versions 5.5 CRACMM2 Input Data -- 12/01/2022 - 12/31/2023 12km CONUS + Canada

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US EPA, 2025, "CMAQ Model Versions 5.5 CRACMM2 Input Data -- 12/01/2022 - 12/31/2023 12km CONUS + Canada", <https://doi.org/10.15139/S3/GL41QC>, UNC Dataverse, DRAFT VERSION 

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Data Summary

CMAQv5.5 input data for the **CRACMM2** chemical mechanism for a 12/22/2021 - 12/31/2022 simulation over the Continental US and Canada.

Note: The datasets are part of the CMAS Center Amazon Web Services (AWS) Open Data Program and are saved in an [AWS S3 bucket](#). The metadata associated with this DOI contain the link to the S3 bucket and instructions for downloading the data. Disclaimer: This data product has been reviewed in accordance with U.S. Environmental Protection Agency policy and approved for publication. Mention of trade names or commercial products does not constitute endorsement or recommendation for use. See below for how to cite this data.

File Location and Download Instructions

- [Link to emissions, initial, and boundary conditions input files](#) for 2023 12US4 CRACMM2
- [Link to download instructions](#)

This S3 buckets contain CMAQ-ready meteorology, emissions, initial, and boundary conditions input files suitable for running a simulation using the CRACMM2 (also compatible with CRACMM3) chemical mechanism. Full documentation on these inputs is available in the Supplemental Information of Pye et al. (2025). A brief overview is presented here.

Meteorology Inputs

Meteorology was generated with WRFv4.6.0 and processed by MCIP resulting in CMAQ-ready files that are located in the **2023_12US4/met** directory. An evaluation of the meteorology, as html pages within a tar file, is available in **2023_12US4/meteval** directory.

Emissions Inputs

Model-ready emissions inputs use CRACMM2 speciation and are available in the **2023_12US4/emis** directory. Files linking representative day emissions to specific calendar days are provided in the **2023_12US4/smk_dates** directory.

2023 CRACMM2 emissions largely follow the [2022 modeling platform inventory methods](#) used to generate the data available from [this AWS S3 bucket](#) except for CRACMM2 VOC speciation (Skipper et al., 2024) and year-specific information when available including for fires. Note CRACMM2 changes how emissions inventories of primary organic aerosol (POA) are allocated to model species of different volatility and functionality. In CRACMM2, POA emissions are assigned to CRACMM species in S2S-Tool+SMOKE and present in the CMAQ-ready files with CRACMM names. The POA inventory is aligned with species of saturation concentrations of 100 µg/m³ and below as described by Pye et al. (2025). DESID should not be used to apply POA volatility profiles to emissions as they have already been applied upstream.

Year-specific retrospective Canada fire emissions inventories used in the ptfire_othna sector emissions inputs are available in the **2023_12US4/inven/ptfire_canada** directory.

Daily county, province, or other jurisdiction emissions of PM_{2.5} from wood combustion in kg/day are provided in csv files available in the **2023_12US4/inven/woodcombustiontotals** directory.

Boundary and Initial Conditions Inputs for 12US4 CMAQ CRACMM2 Simulation

Boundary conditions files are in the **2023_12US4/icbc/** directory.

An initial conditions file generated from typical vertical profile of concentrations distributed with CMAQ BCON is located in the **2023_12US4/initial/CON/v55_2023_12US4_prof2cracmm2** directory.

Boundary conditions were obtained from GEOS-CF and mapped to CRACMM using aqmbc (<https://github.com/barronh/aqmbc>). Data from GEOS-CF was downloaded for every three hours and regridded and mapped to the CMAQ domain.

Initial conditions are based on the profile distributed with CMAQv5.5 and mapped from CB6 to CRACMM using combine and the species definitions files from CMAQ.

Additional required inputs for CMAQ

In addition to these meteorology, emissions, initial, and boundary conditions files, users will also need certain other input files to perform a 2023 12US4 simulation of CMAQv5.5 using CRACMM. The following files are included with this 2023 data repository:

- GRIDDESC grid description file for the 12US4 domain available here: **2023_12US4/GRIDDESC**
- Input files for land surface properties including ocean area for sea spray emissions, land use and other data for biogenic emissions, and Environmental Policy Integrated Climate (EPIC) outputs to drive ammonia bidirectional flux. Files are available in the **2023_12SU4/surface** directory
- Lightning NO emission input files based on lightning data from WWLLN. Files are available in the **2023_12US4/met/lightning** directory.

Scripts and control files

- Inputs here may be used in CMAQv5.5 or later versions with CRACMM2, CRACMM3, or CRACMM3HAPs. CRACMM2 and CRACMM3 emissions are fully cross-compatible.
- For an example build script, run script, and control file for running CRACMM3HAPs see <https://doi.org/10.5281/zenodo.16099070>. The user may need to customize these files to work for other model mechanisms and versions. The run script can be configured for CRACMM3 or CRACMM2 by removing the explicit HAPs (see also <https://github.com/USEPA/CMAQ>).

Resources for setting up a CMAQ simulation

CMAQ benchmark tutorial for CRACMM2: describes how to prepare your Linux system for installing and running CMAQ.

File Format

The 2023 model inputs are stored as uncompressed netcdf formatted files using I/O API data structures. Information on the model projection and grid structure is contained in the header information of the netcdf files. The netcdf files can be opened and manipulated using I/O API utilities (e.g. M3XTRACT, M3WINDW) or other software programs that can read and write netcdf formatted files (e.g. Fortran, R, Python).

CRACMM2 emissions are compatible with the CRACMM3 mechanism

No changes are needed to the DESID emissions rules for switching between CRACMM2 and CRACMM3. Note that the example scripts are for CRACMM3HAPs and some tracer emissions should be removed when using CRACMM2 and CRACMM3 (without HAPs).

Using these emissions in CRACMM1 requires minor adjustments following this guidance: <https://usepa.github.io/CRACMM/emissions/FEA2M1.html>

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Recommended citations

For use of this data, please cite both the dataset for traceability and the manuscript for method documentation:

US EPA, 2025, "CMAQ 2023 12US4 CRACMM2 Inputs and Meteorology", <https://doi.org/10.15139/S3/GL41QC>, UNC Dataverse, V1.

H. O. T. Pye, W. T. Hutzell, N. Fann, T. N. Skipper, M. Pye, J. Beidler, C. Allen, B. Murphy, E. L. D'Ambro, S. Lin, K. Talgo, L. Reynolds, D. Kang, J. Bash, K. Seltzer, S. Farrell, K. W. Appel, K. Brehme, R. Gilliam, B. Henderson, and A. W. H. Chan: The risks to human health of air toxics, PM2.5, and ozone from the 2023 Canadian wildfires, ChemRxiv, 2025. This content is a preprint and has not been peer-reviewed. Please update to the final journal article when available.

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Subject ?

Earth and Environmental Sciences

Keyword ?

2023, CMAQ, CMAQv5.5, 12US4,CRACMM

Related Publication ?

Skipper, T. N., D'Ambro, E. L., Wiser, F. C., McNeill, V. F., Schwantes, R. H., Henderson, B. H., Piletic, I. R., Baublitz, C. B., Bash, J. O., Whitehill, A. R., Valin, L. C., Mouat, A. P., Kaiser, J., Wolfe, G. M., St. Clair, J. M., Hanisco, T. F., Fried, A., Place, B. K., and Pye, H. O. T.: Role of chemical production and depositional losses on formaldehyde in the Community Regional Atmospheric Chemistry Multiphase Mechanism (CRACMM), Atmos. Chem. Phys., 24, 12903–12924, <https://doi.org/10.5194/acp-24-12903-2024>, 2024.

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Files

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Resources

- UNC Dataverse User Guide
- UNC Dataverse Policies
- RDMC Data Services

Support

- Submit via RDMC service portal
- Email RDMCArchive@unc.edu